Politecnico di Torino

Electronic Engineering - Embedded Systems

Lab 4 Report

Numerical analysis of band pass filters

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1 Process description

1.1 Problem nr.1

In problem nr.1 we had to compute the frequency behavior of $|S_{21}\ dB|$ for three different cases. By using the formulas for evaluating ABCD parameters for 2-port tranmission lines, that are:

$$\begin{aligned} A &= \cos\beta\ell, \qquad & B &= jZ_0 \sin\beta\ell, \\ C &= jY_0 \sin\beta\ell, \qquad & D &= \cos\beta\ell \end{aligned} \tag{1}$$

We can use the following formula:

$$S_{21} = \sqrt{\frac{Z_{r1}}{Z_{r2}}} \cdot \frac{2Z_{r2}}{Z_{r2}A + B + Z_{r1}Z_{r2}C + Z_{r1}D} \tag{2}$$

And evaluating the magnitude directly from MATLAB we get the ending result that is:

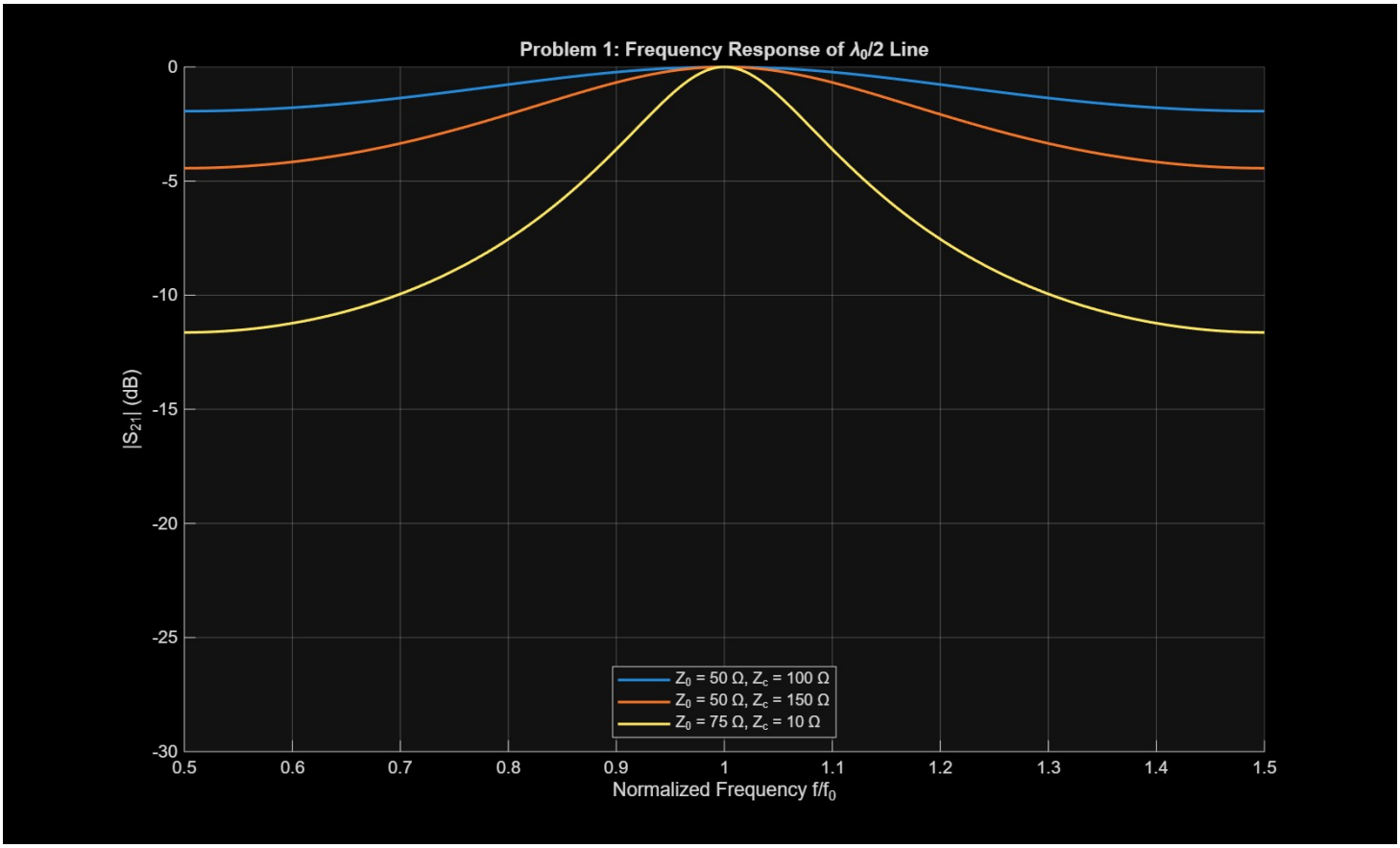

Figure 1: Frequancy response of $\lambda_0/2$ transmission line

As we can appreciate from the graph we can see that at the design frequency f_0 the line is totally neutral which is coherent with the fact that this is a transmission line of $\frac{\lambda_0}{2}$ lenght so it "copies" the impedence at port 1 directly at port 2, without any attenuation. Moving forwards or backwards from the design frequency attenuation increases because

the eletrical lenght of the line, that is strongly dependant on frequency, is not $\frac{\lambda_0}{2}$ anymore. Another thing that can be noted is the fact that junctions at the two ports create reflection spikes, the more the differrence in impedence from Z_0 to Z_c the bigger the attenuation. Last thing, related to the latter comment is that having a line with impedence lower than the impedence at the port creates a "higher" resonance peak, which makes for a better band pass filter.

1.2 Problem nr.2

For problem nr.2 we had to create a filter that respected the following specifications:

$$\begin{aligned}
&\text{Case 1:}\\
&\text{fractional bandwidth: } 30\%;\\
&|S_{21}(f)|_{dB} \text{ at } f = 0.5f_0 \text{ and } f = 1.5f_0 : -20\,\text{dB};\\
&\text{max ripple: } 0.5\,\text{dB}.\\
&\text{Case 2:}\\
&\text{fractional bandwidth: } 40\%;\\
&|S_{21}(f)|_{dB} \text{ at } f = 0.5f_0 \text{ and } f = 1.5f_0 : -35\,\text{dB};\\
&\text{ripple} = 1.5\,\text{dB};\\
&\text{use as a “worst case” the curves for ripple = 0.5 dB.}
\end{aligned} \tag{3}$$

The first thing to do was getting the order of the final filter and this was possible thanks to this data:

$$IL = -S_{21_{dB}} \tag{4}$$

$$\tilde{\omega} = \frac{1}{\Delta} \cdot (\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega}) \tag{5}$$

and the values that we got from the specifications were: For case 1:

$$IL = 20\ dB \tag{6}$$

$$\tilde{\omega_1} = \frac{1}{0.3} \cdot (\frac{2\pi\ \cdot\ 0.5f_0}{2\pi f_0} - \frac{2\pi f_0}{2\pi\ \cdot\ 0.5f_0}) = -5 \tag{7}$$

$$\tilde{\omega_2} = \frac{1}{0.3} \cdot (\frac{2\pi\ \cdot\ 1.5f_0}{2\pi f_0} - \frac{2\pi f_0}{2\pi\ \cdot\ 1.5f_0}) = 2.77 \tag{8}$$

As we can see $\tilde{\omega_2}$ is more restrictive, so we use this one. For case 2:

$$IL = 35\ dB \tag{9}$$

$$\tilde{\omega_1} = \frac{1}{0.4} \cdot (\frac{2\pi \ \cdot \ 0.5f_0}{2\pi f_0} - \frac{2\pi f_0}{2\pi \ \cdot \ 0.5f_0}) = -3.75 \tag{10}$$

$$\tilde{\omega_2} = \frac{1}{0.4} \cdot (\frac{2\pi \ \cdot \ 1.5f_0}{2\pi f_0} - \frac{2\pi f_0}{2\pi \ \cdot \ 1.5f_0}) = 2.08 \tag{11}$$

As we can see $\tilde{\omega_2}$ is more restrictive, so we use this one. By putting these values in the graph for a polynomial approximation and prototype of a filter that has an equal-ripple of 0.5 dB:

$$For\ case\ 1 => N = 3 \tag{12}$$

$$For\ case\ 2 => N = 5 \tag{13}$$

For impedences dimensioning we adopted a naive approach. The base idea was starting with the very first and very last values that the lab3.pdf allows us to use and then change the impedence of the "high impedence" section in order to match the IL requested.

1.2.1 Naive approach: first try

The very first try was made with these values: For case 1:

$$Z_{case\ 1} = [20, 180, 20]; \tag{14}$$

For case 2:

$$Z_{case\ 2} = [20, 180, 20, 180, 20] \tag{15}$$

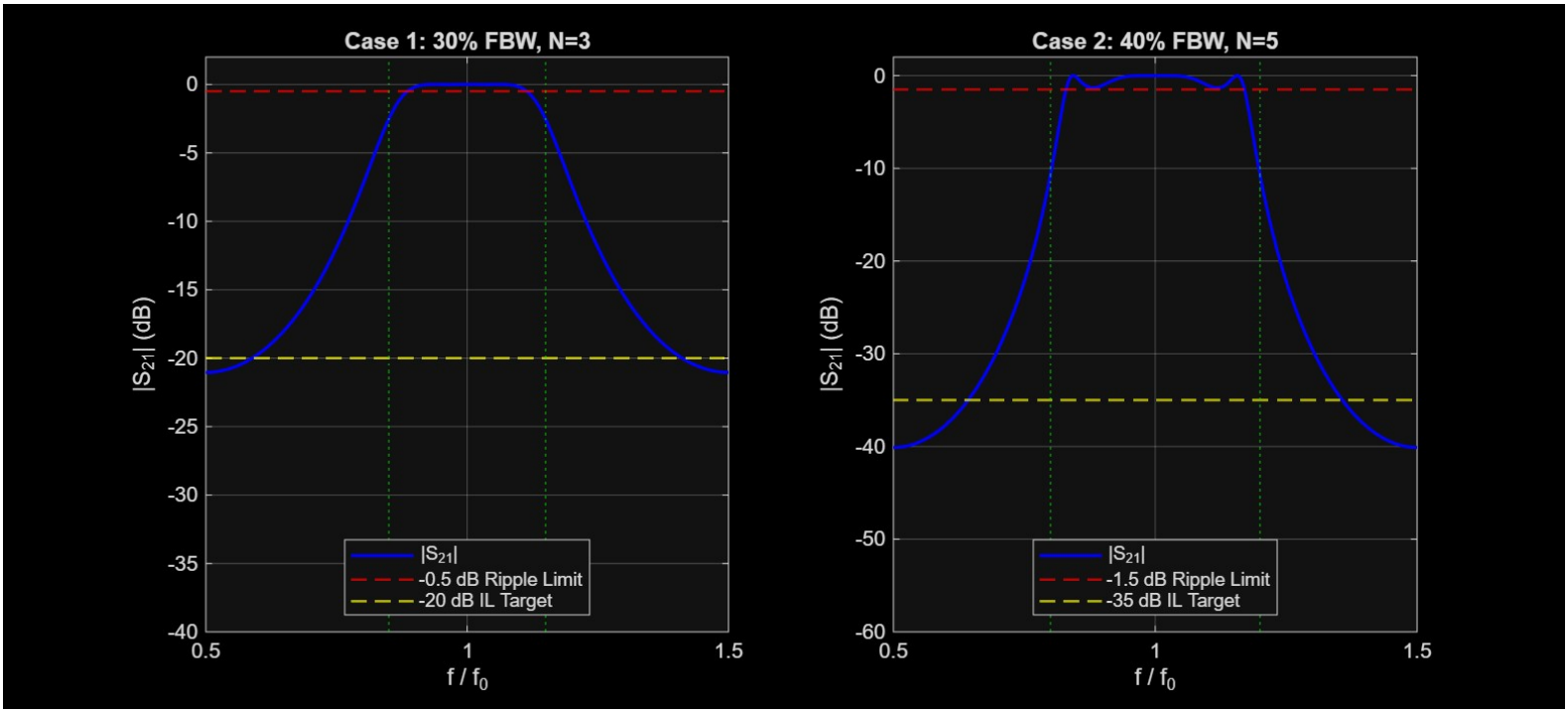

Figure 2: Problem nr.2 naive solution: first try

1.2.2 Naive approach: correct solution

As we can see the constraints on IL in not quite met, so we had to change the values of impedences. For case 1:

$$Z_{case\ 1} = [20, 160, 20]; \tag{16}$$

For case 2:

$$Z_{case\ 2} = [20, 135, 20, 135, 20] \tag{17}$$

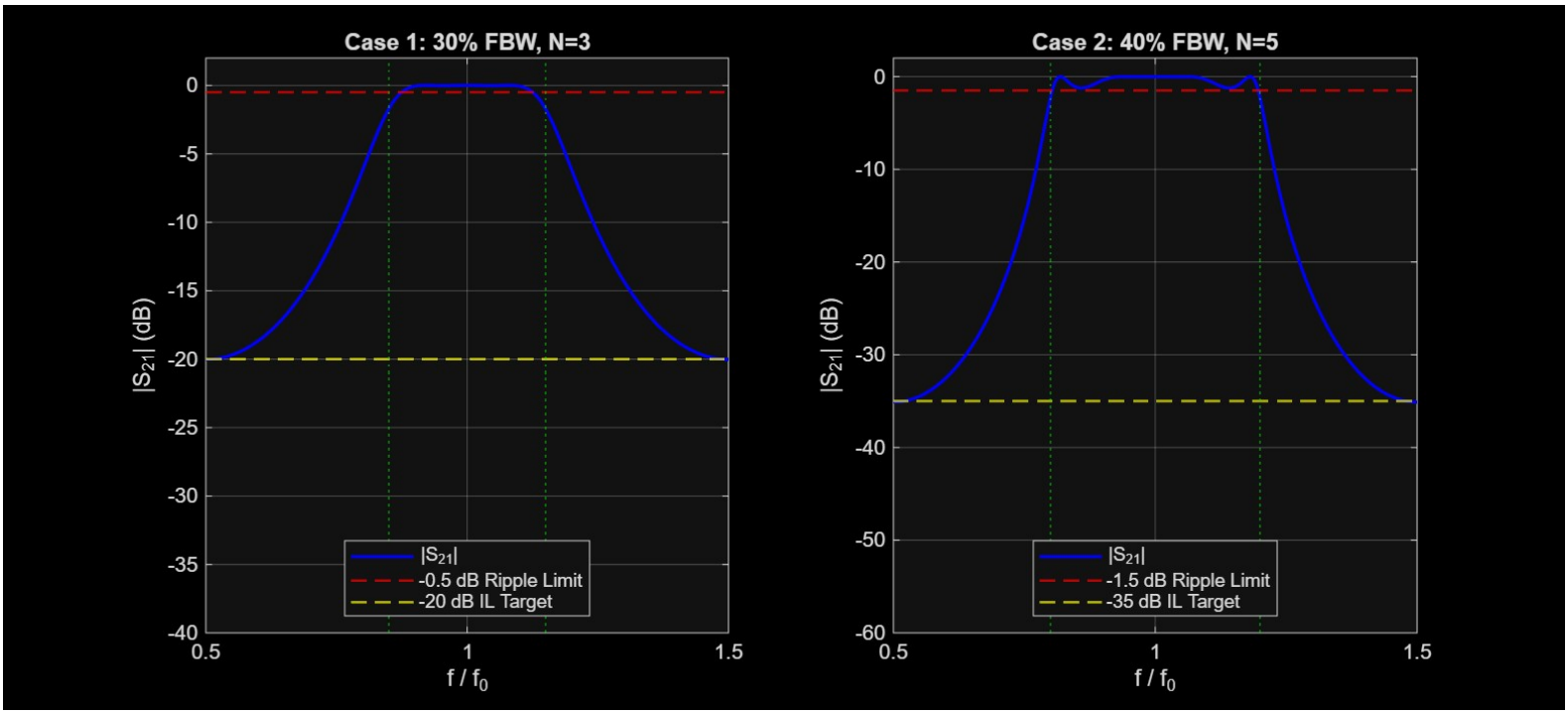

Figure 3: Problem nr.2 case 1 and case 2

As we can see from this graph every specification is met by using these values. The thing that we can notice here is that by having a filtetr of an higher order it means that we can get more attenuation and a more steep rise and descend of the $|S_{21_{dB}}|$ graph and this means that the band pass filter will be more selective.

A Appendix

A.1 Problem nr.1 MATLAB code

```
f_norm = linspace(0.5, 1.5, 500); % Normalized frequency f/f0
beta_l = pi * f_norm;             % Electrical length

% Define the test cases
cases = [50, 100;
```

```
          50, 150;
          75, 10];

figure; hold on;
for i = 1:size(cases, 1)
    Z0 = cases(i, 1);
    Zc = cases(i, 2);

    % We compute ABCD parameters for a transmission line and we use them in the formula to compute
    % S21, at the sime time we calculate the magnitude.
    mag_S21 = 1 ./ sqrt(cos(beta_l).^2 + ((Zc^2 + Z0^2)/(2*Z0*Zc)).^2 .* sin(beta_l).^2);
    S21_dB = 20 * log10(mag_S21);

    plot(f_norm, S21_dB, 'DisplayName', sprintf('Z_0 = %d \\Omega, Z_c = %d \\Omega', Z0, Zc), 'LineWidth', 1.5);
end

grid on;
xlabel('Normalized Frequency f/f_0');
ylabel('|S_{21}| (dB)');
title('Problem 1: Frequency Response of \lambda_0/2 Line');
legend('Location', 'south');
ylim([-30 0]);
```

A.2 Problem nr.2 MATLAB code

```
f_norm = linspace(0.5, 1.5, 500);
beta_l = pi * f_norm; % Electrical length of lambda/2 @ f0
Z0 = 50; % Reference impedance

% Case 1
% TEST CASE 1:
%Zc_case1 = [20, 180, 20];
% CORRECT CASE 1:
Zc_case1 = [20, 160, 20];

% Case 2
```

```
% TEST CASE 2
%Zc_case2 = [20, 180, 20, 180, 20];
% CORRECT CASE 2
Zc_case2 = [20, 135, 20, 135, 20];

% Preallocate arrays for speed
S21_case1 = zeros(1, length(f_norm));
S21_case2 = zeros(1, length(f_norm));

% Calculate Cascaded S21
for k = 1:length(f_norm)
    bl = beta_l(k);

    % --- Case 1 Cascade (N=3) ---
    ABCD_1 = eye(2);
    for i = 1:length(Zc_case1)
        Zc = Zc_case1(i);
        A = cos(bl);
        B = 1i * Zc * sin(bl);
        C = 1i * (1/Zc) * sin(bl);
        D = cos(bl);
        ABCD_1 = ABCD_1 * [A, B; C, D];
    end
    S21_case1(k) = 2*Z0 / (Z0*ABCD_1(1,1) + ABCD_1(1,2) + Z0^2*ABCD_1(2,1) + Z0*ABCD_1(2,2));

    % --- Case 2 Cascade (N=5) ---
    ABCD_2 = eye(2);
    for i = 1:length(Zc_case2)
        Zc = Zc_case2(i);
        A = cos(bl);
        B = 1i * Zc * sin(bl);
        C = 1i * (1/Zc) * sin(bl);
        D = cos(bl);
        ABCD_2 = ABCD_2 * [A, B; C, D];
    end
    S21_case2(k) = 2*Z0 / (Z0*ABCD_2(1,1) + ABCD_2(1,2) + Z0^2*ABCD_2(2,1) + Z0*ABCD_2(2,2));
end
```

```
% --- Plotting ---
figure('Position', [100, 100, 1000, 450]);

% Convert complex S21 to dB
S21_case1_dB = 20*log10(abs(S21_case1));
S21_case2_dB = 20*log10(abs(S21_case2));

% Case 1 Plot
subplot(1, 2, 1);
plot(f_norm, S21_case1_dB, 'b-', 'LineWidth', 1.5); hold on;
yline(-0.5, 'r--', 'LineWidth', 1.2);
yline(-20, 'y--', 'LineWidth', 1.2);
xline(0.85, 'g:');
xline(1.15, 'g:');
ylim([-40, 2]);
title('Case 1: 30% FBW, N=3');
xlabel('f / f_0'); ylabel('|S_{21}| (dB)');
grid on;
legend('|S_{21}|', '-0.5 dB Ripple Limit', '-20 dB IL Target',
    'Location', 'south');

% Case 2 Plot
subplot(1, 2, 2);
plot(f_norm, S21_case2_dB, 'b-', 'LineWidth', 1.5); hold on;
yline(-1.5, 'r--', 'LineWidth', 1.2);
yline(-35, 'y--', 'LineWidth', 1.2);
xline(0.80, 'g:');
xline(1.20, 'g:');
ylim([-60, 2]);
title('Case 2: 40% FBW, N=5');
xlabel('f / f_0'); ylabel('|S_{21}| (dB)');
grid on;
legend('|S_{21}|', '-1.5 dB Ripple Limit', '-35 dB IL Target',
    'Location', 'south');
```